Mehdi Latif
PhD in Signal Processing

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in mehdilatif

My research interests concern the development of algorithmic methods at the interface between *optimization*, *inverse problems* and *probabilistic modeling*.

My PhD work focuses on probabilistic models and iterative tomographic reconstruction methods for *PET/Compton* imaging from complex data. I previously worked on sparse optimization for spectral unmixing, using exact *branch-and-bound* methods.

I wish to pursue my research on numerical methods for inverse problems in signal and image processing, drawing in particular on tools from optimization, inference and simulation, with an outlook open to different fields of application.

Inverse problems | Optimization | Tomographic reconstruction | Statistical inference | Sparsity

Research

2021–2025 **PhD thesis** — École Centrale de Nantes
Tomographic reconstruction for 3-photon imaging based on hybrid PET/Compton cameras.
Supervision : Jérôme Idier , Simon Stute et Thomas Carlier.
PhD carried out within LS2N (SiMS) and CRCI^{2}NA (Nuclear Oncology). Manuscript available online : mlatif.fr.

2020–2021 **Research Engineer** (13 months) — SiMS Team, LS2N, Nantes.
Continuation of the M2 internship work as part of the ANR MIMOSA project : implementation of an exact branch-and-bound algorithm and writing of the associated scientific article.

2020 **Master 2 Internship** (6 months) — SiMS Team, LS2N, Nantes.
MIMOSA UNMIX : Development of a branch-and-bound algorithm for the sparse spectral unmixing problem under non-negativity and sum-to-one constraints.
Supervision : Sébastien Bourguignon and Ramzi Ben Mhenni.

2018 **Bachelor's Internship** (2 months) — MODELIS Team, LS2N, Nantes.
Design of a matheuristic for the pickup and delivery problem with time windows.
Supervision : Fabien Lehuédé and Quentin Tonneau.

Education

2021 – 2025 **PhD, Signal, Image, Vision** — École Centrale de Nantes
Tomographic reconstruction for 3-photon imaging based on hybrid PET/Compton cameras.
Supervision : Jérôme Idier, Simon Stute and Thomas Carlier.

2018 – 2020 **Master's in Computer Science** — University of Nantes / Université libre de Bruxelles
Track : Operations Research Optimization, dual degree with ULB.
Master's thesis : *Graph theory applied to timetable design under assignment constraints*, supervised by I. Rusu.

2015 – 2018 **Bachelor's in Computer Science** — University of Nantes, France
Mathematics track.

2013 – 2015 **Two-year university degree in technology** — IUT de la Fleuriaye, Nantes
Industrial Logistics Quality and Organization major.

2011 – 2013 **Polytech Schools** — University of Tours

Polytech engineering school – preparatory classes :*Mathematics and Computer Science specialization*.

2007 – 2011 **Scientific Baccalaureate** Lycée Clemenceau, Nantes
Mathematics specialization, Arabic option.

Publications

Peer-reviewed international conference proceedings

2025 **Guaranteed Best Sparse Solutions for Spectral Unmixing.**
N.Foix-Colonier, S.Bourguignon, F.Schmidt, M.Latif
WHISPERS, Barcelona, Spain.
hal:05319871

2025 **Branch-and-bound algorithm for exact ℓ_0-norm sparse spectral unmixing.**
M.Latif, G.Samain, N.Foix-Colonier, S.Bourguignon
EUSIPCO, Palermo, Italy.
hal:05124474

2023 **Multi-class maximum likelihood expectation-maximization list-mode image reconstruction, an application to three-gamma imaging.**
M.Latif, J.Idier, T.Carlier, S.Stute
Fully3D, Stonybrook, USA.
hal:04205077

Peer-reviewed national (French) conference proceedings

2025 **Ensemble des solutions parcimonieuses exactes en démélange spectral : algorithme garanti et analyse des solutions.**
N.Foix-Colonier, S.Bourguignon, F.Schmidt, M.Latif
GRETSI, Strasbourg, France
hal:05145133

2023 **Reconstruction multiclasse pour l'imagerie TEP 3-photons.**
M.Latif, J.Idier, T.Carlier, S.Stute
GRETSI, Grenoble, France
hal:04197715

Talks

PhD Defense.

Mar 2026 **PhD Defense**, École Centrale de Nantes.
Tomographic reconstruction for 3-photon imaging based on hybrid PET/Compton cameras.
mlatif:slides mlatif:manuscrit

International Conferences.

Jul. 2023 **17th International Meeting on Fully 3D Image Reconstruction in Radiology and Nuclear Medicine** *(Fully3D),* Stonybrook, USA.
Multi-class maximum likelihood expectation-maximization list-mode image reconstruction, an application to three-gamma imaging.
Peer-reviewed conference.

National Conferences.

Aug. 2023 **XXIXth Francophone Conference on Signal and Image Processing** *(GRETSI),* Grenoble.
Reconstruction multiclasse pour l'imagerie TEP 3-photons.
Peer-reviewed conference.

Invited Seminars and Working Groups.

Nov 2022 **Workshop Emily : Lyon Meeting on Emission Imaging**, Lyon.
Challenges and opportunities of three-photon tomographic reconstruction for the XEnon Medical Imaging System.
mlatif:emily

Oct 2022 **LS2N PhD Students' Half-Hour Talk**, Nantes.
Challenges and opportunities of three-photon tomographic reconstruction for the XEnon Medical Imaging System.

Sep 2021 **SiMS Team Seminar**, Nantes.
Exact resolution of the sparse spectral unmixing problem.
mlatif:unmix

</> Software & Tools

MIMOSA-UNMIX : Branch and Bound for sparse spectral unmixing
C++ implementation of the exact solver for sparse spectral unmixing under non-negativity and sum-to-one constraints.
Code Instances

POLLUX : Monte-Carlo simulator dedicated to 3γ imaging
Written in MATLAB for generating particle-physics data (XEMIS2) ; outputs compatible with CASToR.
Code

ORION : *Observation-based Response Integration for sensitivity estimatiON*
Monte-Carlo generator of CASToR-compatible multiclass sensitivity maps, dedicated to XEMIS2.
Code

ARIANE & PROTEUS : *Analytical Region Identification for Admissible detection in Nuclear Emission & PRojection Operator based on Transformed Elementary Unit Shapes*
Analytical geometric screening algorithm for accelerating the computation of the projection operator in Compton reconstruction, and an analytical operator evaluation method based on deforming voxels into spherical boxes.
Code

CASToR-XEMIS2 : Extension of the CASToR reconstruction software for multiclass 3γ imaging
Integration of list-mode data format for continuous acquisitions, multiclass reconstruction algorithm, and Compton projector.
Code

Science Outreach

Feb 2024 **INPACT Seminar – Art and Science,** Nantes.
Presentation of research work as part of the professional track *Engineering and Digital Technology for Heritage, Art and Culture* (École Centrale de Nantes).
mlatif:impact

2023, 2024 **Researchers' Night,** Nantes.
Science-dating : presentation of the PhD project in under 7 minutes.

Teaching

École Centrale de Nantes (teaching assignments) ∼**89 TD-equivalent hours** (French teaching-load unit)

2024-2025 **Teaching Assistant** (labs and grading)
- *Scientific Computing and Optimization*, 2nd-3rd year Engineering Students, DATASIM Track (8h) ;
- *Mathematical Tools for Signal and Image Processing*, Master 2 CORO, track DASSIP (4h).

2023-2024 **Teaching Assistant** (labs and grading)
- *Scientific Computing and Optimization*, 2nd-3rd year Engineering Students, DATASIM Track (12h) ;
- *Mathematical Tools for Signal and Image Processing*, Master 2 CORO, track SIP (4h) ;
- *Mathematics for Engineers*, 1st year Engineering Students (24h) ;
- *Signals and Systems*, 1st year Engineering Students (10h).

2022-2023 **Teaching Assistant** (labs and grading)
- *Mathematics for Engineers*, 1st year Engineering Students (24h) ;
- *Signals and Systems*, 1st year Engineering Students (4h) ;
- *Algorithms and Programming*, 1st year Engineering Students (18h).

2021-2022 **Teaching Assistant**
- *Algorithms and Programming*, 1st year Engineering Students (18h).

Student Supervision

2023 **Master 2 Literature Review Project** (3 months)
Y. Oumarou, international Master 2 CORO-SIP (ECN).
ComProj project : State of the art of projectors in Compton imaging.

2023 **Master 1 Internship** (4 months)
E. Gecimli, Master 1 Fundamental Physics and Applications (University of Nantes).
Study and implementation of event-reconstruction algorithms for 3-photon PET imaging applied to the XEMIS2 camera. Extraction and analysis of data from a GEANT4 simulation of the XEMIS2 camera, in collaboration with N. Beaupère (Subatech Laboratory).

Professional Experience

2015–2016 **PEPITE Créer Nantes** (1 year) Nantes Université
National student-entrepreneur status – Veckevent project.
Recipient of the Passeport Armorique pour Entreprendre award ; managed a team of 11 students.

2015 **Industrial Improvement Internship** (3 months) Groupe Pilote SAS-CSA Gérard, Sainte-Luce-sur-Loire.
Development and implementation of logistics methods in a craft-based production environment.

2013 **Industrial Production Internship** (1 month) Airbus Operations SAS, Nantes.
Introduction to composites-unit activities, with immersion in quality, logistics and production support.

2013 **Web Development Internship** (1 month) Dadeyo, Nantes.
Development of a Kanban Board web application.

2012 **Networks and Systems Internship** (1 month) Orange Business Services, Nantes.
Introduction to network operations activities and work on application probes.

Skills

Technical Skills

- Windows MacOS Linux
- Bash C++ Git Matlab R
- LaTeX Beamer Office
- GIMP Inkscape TikZ

Languages

French	●●●●●
English	●●●●○
German	●○○○○

Interests

Arts	Photography, literature, music (music theory, French horn, guitar).
Sports	Rowing, badminton.
Other	Travel (Europe, North America, North Africa).